
S/4HANA Clean Core Modernization

A free, non-commercial community pilot project for automated legacy ABAP refactoring to SAP BTP CAP (Node.js/TS) and Developer Extensibility (ABAP Cloud RAP).

01. What is Clean-Core.io?

In simple terms, Clean Core means keeping the standard SAP ERP core code completely untouched. When custom code is mixed directly with standard standard classes and tables, future S/4HANA system updates become complex, slow, and expensive.

This is exactly where **Clean-Core.io** comes in. Designed as a free, non-commercial DIY community learning project, this tool supports SAP developers and architects as an intelligent co-pilot. Simply upload your legacy, proprietary ABAP code (such as raw database SELECTs or obsolete function modules) and transform it into modern, upgrade-safe cloud-ready extensions with a single click.

THE PROBLEM: Z-TABLE CHAOS & RIGID CORE MODIFICATIONS

Historically grown SAP systems often contain hundreds of custom database tables and direct modifications. These Z-objects block standard upgrades because SAP can modify the underlying standard structures at any time.

The solution is a clean separation of custom logic and standard software. Our wizard analyzes your legacy Z-objects and automatically recommends the optimal modernization track.

S/4HANA Extensibility Rule: Custom developments belong either as In-App Extensibility directly in the SAP ABAP Cloud Stack (RAP) or as Side-by-Side Extensibility on the SAP Business Technology Platform (BTP CAP). The standard core remains completely clean and untouched.

02. The Value: What benefits does it bring to the user?

Clean-Core.io is not just a code translator, but a comprehensive modernization accelerator. Developers, system architects, and IT decision-makers benefit from immediate, tangible advantages in daily operations:

ENORMOUS TIME SAVINGS

Manually searching for modern SAP standard APIs and rewriting legacy ABAP into modern Node.js microservices or RAP classes takes days. Clean-Core.io executes the analysis and code generation in under 2 minutes.

UPGRADE RESILIENCE

By replacing unreleased database dependencies with officially released standard SAP APIs (like I_Customer or API_PRODUCT_SRV), your ERP core remains fully upgradable and future-proof.

AUTOMATED TEST STUBS

The tool writes matching automated unit tests immediately – as ABAP Unit Test Doubles for RAP or as Playwright/Express testcases for CAP. This saves extensive QA cycles.

BPMN 2.0 SIGNAVIO EXPORT

L3 business processes are exported in standard BPMN 2.0 XML schema. These files can be directly imported into SAP Signavio Process Manager or SAP Build to visualize your processes instantly.

THE FAST-PASS ADVANTAGE

Using the integrated Fast-Pass Stepper, you can bypass intermediate wizard stages and jump straight from the analysis page to the Delivery view to download your compiled ZIP package containing all modularized project files.

03. Closed Pilot, Data Security & Compliance

Proprietary legacy source code is a highly confidential business asset. Clean-Core.io is engineered with strict cryptographic and operational safety boundaries to ensure data compliance and protection.

THE CLOSED PILOT: CONTROLLED ACCESS & BYOK

The platform is accessible via a restricted, free Closed Pilot. Every new pilot user receives 5 free transformations to explore the wizard.

For unlimited, private, and fully secure enterprise modernizations, the platform supports the innovative **Bring Your Own Key (BYOK)** vault. Your private Gemini API key is encrypted directly inside your browser using PBKDF2 (100,000 iterations) and AES-GCM 256-bit encryption. Plaintext keys never reach our database.

STATELESS AI ROUTING & MODEL TRAINING ISOLATION

All AI prompts are routed via a secure, server-side API proxy to Google Vertex AI.

In accordance with Google Cloud Enterprise data safety terms, all code snippets sent to the backend proxy are **strictly excluded from LLM model training and reinforcement cycles**. Your intellectual property remains strictly yours.

100% GDPR COMPLIANCE & SOVEREIGNTY (ART. 25 & 17 GDPR)

All server, database, and compute layers are hosted entirely within the GCP region **Belgium (europe-west1)**, satisfying European data residency policies.

Triggering account deletion initiates an atomic backend routine (Art. 17 GDPR - Right to Erasure) that permanently wipes Firebase Auth records, Firestore project documents, and stored ZIP files instantly. The system never parses or processes actual business master data.

04. Technical Details: RAP vs. CAP & API Mapping

The modernization engine decides during Stage 1 based on syntax analysis which extensibility path is best suited for your legacy system:

TECHNICAL DIMENSION	IN-APP ABAP CLOUD (RAP)	SIDE-BY-SIDE (BTP CAP)
Runtime Engine	Runs natively directly within the S/4HANA core.	Runs decoupled on the SAP BTP (Node.js/TS).
Interfaces	Uses synchronous released CDS views.	Decoupled via OData APIs or Event Mesh.
RISE Compliance	Strict SaaS compliance (zero core modifications).	Upgrade-resilient Tier 2 custom API wrappers.
Focus Case	Immediate database updates and transactional locks.	Customer portals, external mobile apps, SaaS tools.

AUTOMATED API MAPPING & DECOUPLING

Direct SELECTs on old tables (like KNA1 for customers or BSEG for financials) trigger immediate audit errors. The engine maps these queries to modern standard interfaces:

- ✓ KNA1 (Customer Master) → Maps to CDS View I_Customer
- ✓ BSEG (Accounting Segment) → Maps to CDS View I_JournalEntry
- ✓ MARA (Material Master) → Maps to OData API API_PRODUCT_SRV
- ✓ VBAK (Sales Header) → Maps to RAP Entity I_SalesOrderTP

05. Quality Engineering & Community Collaboration

A clean refactoring is only complete when the result is fully functional and auditable. Clean-Core.io equips every transformation with built-in quality assurance:

AUTOMATED SANDBOX TESTS

For BTP CAP projects, an Express unit test suite is generated. By inputting a BTP Destination JSON, you can securely execute sandbox tests against your non-productive S/4HANA tenant via the SAP Cloud Connector.

ABAP UNIT DOUBLES

For RAP blocks, a local ABAP Unit test class is generated. Database SELECTs are simulated using local doubles, allowing you to run your validation classes immediately and locally.

JOIN THE CLEAN-CORE COMMUNITY

Bring your legacy Z-programs into the cloud era. Register for the Closed Pilot or leverage your own Gemini API key (BYOK) for unlimited, private transformations today!

REGISTER NOW & START → [HTTPS://CLEAN-CORE.IO](https://clean-core.io)

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